

147303 - Signals and systems

UNIT I

1. Determine whether the following systems are linear,time invariant,causal ,stable.

$$(1) \quad y(n) = \log[x(n)]$$

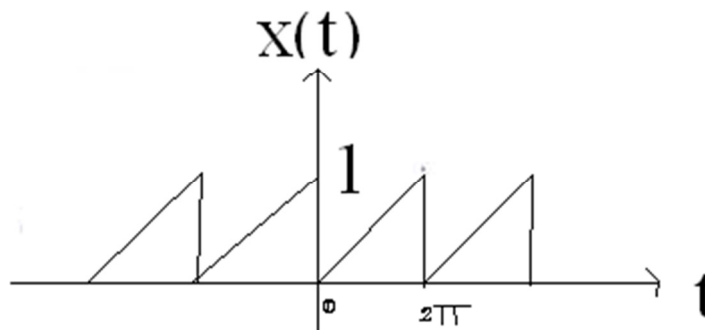
$$(2) \quad y(n) = \sum_{n=0}^{\infty} x(n)$$

2. Determine whether the following systems are linear or not
 $dy(t) / dt + 3ty(t) = t^2x(t)$ & $y(n)=2x(n)+ 1 / x(n-1)$
3. Explain the classification of signals with examples
4. Determine whether the following systems are Time-Invariant or not
 $Y(t) = t x(t)$ & $y(n) = x(2n)$
5. (a) Find whether the signal $x(t) = 2 \cos (10 t+1) - \sin(4t-1)$ is periodic or not.
(b) Evaluate $\sum_{n=(-\infty \text{ to } \infty)} e^{2n} \delta (n-2)$
(c) Find the fundamental period of the Continuous time signal

$$x(t) = 20 \cos \left(10 \pi t + \frac{\pi}{6} \right).$$

UNIT II

1. Find the inverse laplace transform of $X(S) = S / S^2+5S+6$
2. Find the fourier transform of a rectangular pulse of duration T and amplitude A
3. Obtain the cosine fourier series representation of x(t)



$$dy(t) / dt + 5 x(t) = 3 x(t)$$

2. Prove any 2 properties of Z-transform
3. Obtain the cascade form realization of the system described by the differential Equation

$$y(n) - \frac{1}{4} y(n-1) - \frac{1}{8} y(n-2) = x(n) + 3 x(n-1) + 2 x(n-2)$$

4. Find the state variable matrices A,B,C,D for the equation
$$y(n) - 3y(n-1) - 2y(n-2) = x(n) + 5 x(n-1) + 6 x(n-2)$$
5. Discuss the block diagram representation of an LTI-DT system